Fiscal Year:	FY 2022	**Task Last Updated:**	FY 02/04/2022
PI Name:	Patla, Bijunath Ph.D.		
Project Title:	NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	303-497-4082
Organization Name:	National Institute of Standards and Technology		
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City:	Boulder	**State:**	CO
Zip Code:	80305	**Congressional District:**	2
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	04/01/2021	**End Date:**	09/30/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	New Principal Investigator as of April 1, 2021 is Dr. Bijunath R. Patla; previous PI was Dr. Chris Oates.		
COI Name (Institution):			
Grant/Contract No.:	NNH12AT81I		
Performance Goal No.:			
Performance Goal Text:			

NOTE: Continuation of "NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado (PI=Oates)" with new PI, Dr. Bijunath R. Patla.
To support primary Atomic Clock Ensemble in Space (ACES) mission objectives, the National Institute of Standards and Technology (NIST) proposes to:

1. Develop a Microwave Link Ground Terminal (MWL GT) at NIST in Boulder, CO to perform 10^-17 uncertainty time and frequency comparisons between ACES clocks and world-leading optical and atomic clocks at NIST Boulder and the nearby JILA (formerly known as the Joint Institute for Laboratory Astrophysics); and to support general ACES science goals.

2. Participate as fully as possible in coordinated European Space Agency (ESA) data analysis with formal ESA Data

Task Description:	Analysis Centers and other ACES partners throughout the world. NIST will collaborate closely with ESA partners to define how NIST can best contribute to overall ACES mission goals within the constraints of NIST resources. As part of participation in the time and frequency transfer and data analysis, NIST will participate in tests of fundamental physics based on ACES data to include at least the following: • Global time and frequency transfer with uncertainty as low as 10^-17. • Direct comparison of PHARAO space clock to NIST time and frequency standards at uncertainty of 10^-17 or better. • Test of gravitation redshift at 2 x 10^-6 uncertainty or better. • Search for variation in fundamental constants, such as fine structure constant at level of 10^-18 per year or better. All these goals will be facilitated by the strong time and frequency infrastructure at NIST, including (a partial list): • World-class cesium microwave primary frequency standards, including NIST-F1 with fractional frequency uncertainty of 3 x 10^-16, and NIST-F2 which will be operational in 2013 with an eventual uncertainty of 1 x 10^-16 or better. • Two aluminum ion “logic” optical clocks with current fractional frequency uncertainty of approximately 9 x 10^-19. • Two ytterbium lattice optical clocks with uncertainties better than 2 x 10^-18. • Strontium lattice optical clocks with uncertainties of 2 x 10^-18 (at JILA). A dedicated fiber optic system connects NIST and JILA to enable the JILA clocks to directly participate in NIST time and frequency transfer and intercomparisons. • A high performance calcium optical frequency standard with exquisite short-term stability to serve as a “fly wheel” for clock comparisons with demonstrated instability below 10^-16. • A world-class time scale comprising eight hydrogen masers and a collection of Cs clocks. NIST scientists will participate in ESA/NASA planning meetings about the ACES program in general, and will specifically consult with ESA to determine how NIST can most fully participate in MLW GT data analysis and applications. In addition to the NIST resources already proposed for this program (NIST and JILA optical and microwave frequency standards, NIST time scale, NIST time and frequency transfer capabilities, etc.) NIST will directly contribute substantial funds and expert NIST staff time significantly exceeding the direct NASA investment. We propose the following revised program plan outline: Year 1 (2012): Begin design and construction of platform and other infrastructure for MWL hardware. $150k for partial payment of platform and related infrastructure design and construction. Year 2 (2013): Complete construction of platform and other infrastructure for MWL system. $150k for partial payment of platform and related infrastructure. Year 3 (2014): Receive MWL hardware. Install, test, and integrate MWL hardware into NIST time and frequency transfer system. Prepare and test MWL system for ACES time and frequency transfer. New NIST staff to prepare for participation in MWL data analysis. $150k for partial support of staff time. Year 4 (2015): ACES launch. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time. Year 5 (2016): On-going ACES experiments. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time. Gap Year (2017): Perform relativistic calculations to prevent unanticipated shifts from contaminating ACES data. Install updated MWL hardware into new NIST time and frequency transfer system. Prepare updated data analysis system in anticipation of ACES launch. $150k for partial support of staff time. Gap 6 Mo. (2018): Develop and install precision time links between the MWL hardware site and NIST clocks. $80k for partial support of staff time. Gap 12 Mo. (2018-9): Assuming successful launch in 2020, will be prepared to participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. Develop a frequency-based time scale for ACES frequency comparisons. $80k for partial support of staff time. Gap 12 Mo. (2019-2020): Assuming successful launch in 2021, will be prepared to participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. Evaluate and improve the performance of the new frequency-based time scale optimized for ACES frequency comparisons. $80k for partial support of staff time. Gap 12 Mo. (2020-2021): Evaluate and improve the performance of the new frequency-based time scale. Upgrade hardware infrastructure to add resiliency and robustness to the frequency-based time scale. $80k for partial support of staff time.
Rationale for HRP Directed Research:	

Research Impact/Earth Benefits:	The major benefits and impacts of this research program can be grouped into three main areas: 1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements. 2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory. 3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, such as search for possible Lorentz transformation variations and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosmology.
Task Progress:	In the most recent reporting period (October 2020 through January 2022) work has focused on two main tasks: 1. Develop and integrate methods that help with frequency transfer using accurate clocks in space. 2. Improve and evaluate high-precision timing link between the to-be-installed Microwave Link Ground Terminal (MWL GT) and NIST's Time Scale, UTC(NIST). Progress on each task is described below. 1. Develop and integrate methods that help with frequency transfer using accurate clocks in space. We have renewed our efforts for evaluating the fractional frequency shifts of clocks in space due to the higher-order Earth multipole moments. In an inertial reference plane, the orbital elements--right ascension of the ascending node and argument of periapsis are undefined when the inclination of the orbit and the eccentricity are zero. For accurately estimating the higher-order moments using classical perturbation theory, it is more convenient to use an equinoctial coordinate system in which the orbit plane is the reference plane instead of the ecliptic. To this end, we developed and refined a formalism that converts independent variables (orbital elements) in the classical equation of motion to equinoctial variables. With this, one can effectively invoke classical perturbation theory to simplify the calculations for deriving the fractional frequency shifts due to higher-order Earth multipole moments for orbiting clocks. Simulated data sets were generated for clocks in orbits using publicly available orbit determination and analysis software to test and validate several test case scenarios and compare our theoretical framework. In addition to this--to address critical factors that determine the success of any future clock missions in space--we seek answers to the following questions: What are the hurdles to precise and accurate time transfer using optical clocks in space (and ground)? What are the various relativistic effects (and to what higher-order it is to be estimated) for it to matter--for a given measure of clock stability? What are the non-relativistic systematic and statistical effects that need consideration? Continuing the work with the derivation of the transformations from local Earth-centered inertial (ECI) coordinates to local freely-falling coordinates fixed in an orbiting spacecraft, we have added a derivation of the Shapiro time delay from spacecraft to ground, as well as a derivation of light bending. These results are also applicable to any future missions involving clocks in space. 2. Improve and evaluate high-precision timing link between the to-be-installed Microwave Link Ground Terminal (MWL GT) and NIST's Time Scale, UTC(NIST). The frequency scale algorithm continues to be refined and updated. Ongoing tests include adding steers to the source maser to match its output frequency with TAI (temps atomic international), adding and dropping clocks to the ensemble. We demonstrate the robustness of our algorithm by applying large steers while keeping the output frequency of the scale smooth and continuous. High-performance H-masers are now part of the primary time scale, backup time scale, and frequency scale in the same building (hosting the microwave link) as common clocks. We upgraded the measurement systems to accommodate 32 channels from 16 that were supported previously. With multiple links and common clocks, we were able to attain time stability of ~20 ps a day, satisfying a key requirement for the experiment. In the past year, by operating an optical clock network comprising 27Al+, 87Sr, and 171Yb clocks using optical fibre and free-space link, NIST reported measuring their frequency ratios with fractional uncertainties below 10^-17. In addition to redefinition of the SI (International System of Units) second, referencing optical transitions, these measurements demonstrate the potential for testing fundamental physics, and in advancing relativistic geodesy.
Bibliography Type:	Description: (Last Updated: 02/03/2026)

Articles in Peer-reviewed Journals	Boulder Atomic Clock Optical Network (BACON) Collaboration. Beloy K, Bodine MI, Bothwell T, Brewer SM, Bromley SL, Chen J-S, Deschenes J-D, Diddams SA, Fasano RJ, Fortier TM, Hassan YS, Hume DB, Kedar D, Kennedy CJ, Khader I, Koepke A, Leibrandt DR, Leopardi H, Ludlow AD, McGrew WF, Milner WR, Newbury NR, Nicolodi D, Oelker E, Parker TE, Robinson JM, Romisch S, Schaffer SA, Sherman JA, Sinclair LC, Sonderhouse L, Swann WC, Yao J, Ye J, Zhang X. "Frequency ratio measurements at 18-digit accuracy using an optical clock network." Nature. 2021 Mar 25;591(7851):564-9. https://doi.org/10.1038/S41586-021-03253-4 , Mar-2021
Articles in Peer-reviewed Journals	Leopardi H, Beloy K, Bothwell T, Brewer SM, Bromley SL, Chen J-S, Diddams SA, Fasano RJ, Hassan YS, Hume DB, Kedar D, Kennedy CJ, Leibrandt DR, Ludlow AD, McGrew WF, Milner WR, Nicologi D, Oelker E, Parker TE, Robinson JM, Romisch S, Sherman JA, Sonderhouse L, YAo J, Ye J, Zhang X, Fortier TM. "Measurement of the 27Al+ and 87Sr absolute optical frequencies." Metrologia. 2021 Feb;58(1):015017. https://doi.org/10.1088/1681-7575/abd040 , Feb-2021